

NGOC MINH TRAN (PhD)

Dublin, Ireland | +353892493332 | minhitbk@gmail.com | [[LinkedIn Profile](#)] | [[Google Scholar](#)]

PROFESSIONAL SUMMARY

AI Leader with 12+ years of international experience driving enterprise-scale AI Factory initiatives. Currently leading Mastercard's flagship "big bet" project: building and scaling a transaction-based Transformer Foundation Model across hundreds of billions of financial records using H100 GPU clusters. Proven track record in translating deep-tech infrastructure into high-impact banking use cases (SME growth, risk, churn). Expert in AI Security, Federated Learning, and AutoAI with 21 US/EU patents and 16 scientific publications.

CORE COMPETENCIES

- **Generative AI & Infrastructure:** Transformer Architectures, Foundational Models, Large-Scale Distributed Training (H100 Clusters), MLOps at Scale.
- **Banking AI Downstream Use Cases:** Hidden SME Discovery, SME Credit Risk Score, Customer Churn Optimization, Attrition & Reactivation, Expenditure Forecasting, Customer Segmentation.
- **AI Security:** Privacy-Preserving AI, Federated Learning, Adversarial AI.
- **Big Data Engineering:** Enterprise Cloud-Native Infrastructure (Databricks, AWS), Spark, Hadoop.

PROFESSIONAL EXPERIENCE

MASTERCARD | Dublin, Ireland

Director of Data Science / Principal Data Scientist (AI/ML) | 2021 – Present

- **Lead a flagship enterprise "big bet" project** to architect and build a proprietary transaction-based Foundation Model using Transformer architectures.
- **Scale deep learning training** across hundreds of billions of global financial transactions utilizing multi-node H100 GPU clusters.
- **Develop / support downstream models** directly driving banking KPIs, including SME CRS, Hidden SME Identification, Fraud Detection, Customer Attrition, Adoption or Reactivation, Segmentation, and Cardholder Expenditure Forecasting.
- **Optimize multi-million dollar compute budgets** and MLOps pipelines for ultra-large-scale foundational model deployment.
- **Advise executive leadership** on technology investments, scalable AI commercialization roadmaps.

IBM RESEARCH LAB | Dublin, Ireland

Staff Research Engineer (AI & Cloud) | 2017 – 2021

- **Pioneered production-grade frameworks** for Federated Learning and AutoAI tailored for highly regulated cloud and edge banking environments.
- **Led R&D in Enterprise AI Security**, protecting deep learning models against vulnerabilities, adversarial attacks, and data leakage.
- **Secured 4 Outstanding Technical Achievement Awards and Division Awards** for scaling Federated AI and Security AI systems.
- **Invented and filed 21 US & EU patents** as lead/co-inventor, converting deep-tech research into commercial corporate IP portfolios.

SILICON VALLEY STARTUP | California, USA (Remote/Onsite)

Senior AI & Big Data Engineer | 2014 – 2016

- **Architected end-to-end Big Data pipelines** using Apache Spark, Hadoop, and HBase to process petabyte-scale data streams.
- **Developed a cloud-based recommendation engine** for TV, movie, and sports programming serving ~40 million users.
- **Developed IoT machine learning models** for predictive and commercial applications within manufacturing sectors.

EDUCATION

- **Postdoc in Computer Science** | Institute for Research in Computer Science and Automation (INRIA), France | 2012 – 2013
- **PhD. in Computer Science** | Leiden University, The Netherlands | 2008 – 2012

HONORS & ACHIEVEMENTS

Corporate Awards

- [IBM Research Division Award in Security AI 2019.](#)
- [IBM Research Division Award in Auto AI 2019.](#)
- [IBM Outstanding Technical Achievement Award in Security AI 2019.](#)
- [IBM Outstanding Technical Achievement Award in Federated AI for Hybrid Cloud and Edge 2021.](#)
- [IBM First Plateau Invention Achievement Award 2019.](#)
- [IBM Second Plateau Invention Achievement Award 2019.](#)
- [IBM Third Plateau Invention Achievement Award 2020.](#)

Publications & Patents

- **Patents:** Lead inventor (12) and co-inventor (9) of **21 US and EU patents** spanning Security AI, Federated AI, Auto AI, Generative AI, and FinTech AI.
- **Publications:** First author (10) and co-author (6) of **16 scientific papers** in top-tier AI and Cloud Computing venues (Full list via [Google Scholar](#)).

Open Source Recognitions

- [Open Source Strategic Committer in 2019.](#)
- [Pull-Shark achievements 2 times in Github.](#)
- [Arctic Code Vault contributor in Github.](#)

Professional Certifications

- [Developer Certification for Apache Spark](#), certified by Databricks and O'Reilly Media, Jan 2016.
- [Oracle Certified Professional. Java SE 6 Programmer](#), Oct 2015.
- [Revolution R Enterprise Certified Specialist](#), Aug 2015.
- [Cloudera Certified Specialist in Apache HBase](#), Dec 2014.
- [Cloudera Certified Developer for Apache Hadoop](#), Oct 2014.
- [Cloudera Certified Administrator for Apache Hadoop](#), Oct 2014.